

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
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| 4(a)(i)   | RNA polymerase ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>1</b> |
| 4(a)(ii)  | <p>any <b>two</b> from:<br/> <i>accept for either cap or poly(A) tail or both</i></p> <p><b>1</b> protects mRNA from being, broken down / degraded / damaged, by enzymes ;<br/> <b>A</b> protects mRNA from enzyme action</p> <p><b>2</b> prevents mRNA molecules joining together ;<br/> <b>A</b> prevents ends of a mRNA molecule joining</p> <p><b>3</b> helps to, direct / move, mRNA, through nuclear pores / to ribosome(s) / to leave nucleus / to cytoplasm ;</p> <p><b>4</b> required to start, translation / assembly of amino acids at ribosome ;<br/> <b>A</b> helps / allows / AW, (mRNA) attachment to ribosome(s)<br/> <b>R</b> refs to start and stop codons</p> <p><b>5, 6</b> AVP ;; e.g. makes sure 5' end enters ribosome first / AW</p> | <b>2</b> |
| 4(a)(iii) | <p>any <b>three</b> from:</p> <p><b>1</b> gene / RNA, splicing ;<br/> <b>A</b> primary transcript splicing<br/> <b>R</b> DNA splicing / mRNA splicing / genetic splicing</p> <p><b>2</b> introns removed ;</p> <p><b>3</b> exons, attached together / joined up ;<br/> <b>R</b> exons</p> <p><b>4</b> shortens / AW, the RNA molecule ;</p> <p><b>5</b> removal of non-coding sequences / only keeping the coding sequences ;<br/> <b>A</b> non-coding regions / AW<br/> <b>A</b> non-coding introns<br/> <b>R</b> non-coding / coding, genes</p> <p><b>6</b> AVP ; e.g. rearranging exons / alternative splicing <b>I</b> annealing<br/> <i>ref. to phosphodiester bonds forming between RNA nucleotides / AW</i><br/> <i>ref to spliceosome</i></p>        | <b>3</b> |
| 4(b)(i)   | <p>(at the) ends / AW ;<br/> <b>A</b> at the ends of, DNA / chromatids<br/> <b>I</b> sides / edges</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>1</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
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| 4(b)(ii) | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 allows <u>DNA replication</u> to, occur many times ;</li> <li>2 allows (some) cells to carry out, many / continuous / repeated, mitoses / cell cycles / cell divisions ;<br/>I cell replication</li> <li>3 prevents the loss of, genes / genetic information ;<br/>A prevents loss of coding sequences<br/>I loss of DNA<br/>I prevents loss of genetic material</li> <li>4 AVP ; e.g. prevents fusion of chromosome ends<br/>prevents ends of chromosomes being recognised as damaged</li> </ol>                                                                                                                                                                                                                                                                                                                                                | 3     |
| 4(c)     | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 mutation in a gene ;</li> <li>2 <i>leads to</i> uncontrolled / unregulated, mitosis / cell division ;</li> <li>3 proto-oncogene to oncogene ;<br/>A <i>ref. to</i> oncogene in correct context (without the proto-)</li> <li>4 <i>ref. to</i> tumour suppressor gene(s) ;<br/>A in context of 'switched off'</li> <li>5 <i>idea that</i> normal (named) checkpoints do not function ;<br/>A not checked during (named) stage of cell cycle<br/>A bypass checkpoints / checkpoints not used</li> <li>6 mass of, abnormal / non-functional / damaged, cells formed ;<br/>A irregular / abnormal, mass of cells<br/>I undifferentiated</li> <li>7 AVP ; e.g. supplied with blood vessels<br/>cells do not carry out, apoptosis / programmed cell death<br/>cancer cells ignore stop signals from other cells / have no contact inhibition</li> </ol> | 4     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks    |
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| 4(d)     | <p><i>any <b>five</b> from:</i></p> <p><i>melanoma cell releases cytokines</i></p> <p><b>1</b> cytokines stimulate, clonal expansion / division (by mitosis), of, T-lymphocytes / B-lymphocytes ;<br/> <b>A</b> T-cells / B-cells<br/> <b>A</b> T-helper cells / T-killer cells</p> <p><b>2</b> cytokines stimulate (action of), macrophages / phagocytes ;<br/> <b>A</b> form angry macrophages</p> <p><b>3</b> cytokines act as cell-signalling molecules ;</p> <p><i>melanoma cell releases antigens</i></p> <p><b>4</b> antigens stimulate, clonal selection of (specific) B-lymphocytes / T-lymphocytes ;<br/> <b>A</b> B- / T-, lymphocytes (with receptors / immunoglobulins / antibody) that are complementary to antigen<br/> <b>A</b> T-cells / B-cells<br/> <b>A</b> T-helper cells / T-killer cells</p> <p><b>5</b> antigens, stimulate / AW, B-lymphocytes to divide (by mitosis) to form plasma cells ;<br/> <b>A</b> clonal expansion of B-lymphocytes to form plasma cells<br/> <b>A</b> if cytokines stated instead of antigens</p> <p><b>6</b> plasma cells, secrete / release / produce, antibodies ;</p> <p><b>7</b> antibodies mark cancer cells for destruction by, T-killer cells / macrophages / phagocytosis ;</p> <p><b>8</b> killer cells, release / AW, perforin / granzymes / toxins / hydrogen peroxide / hydrolytic enzymes, to kill / destroy, (cancer) cells ;<br/> <b>A</b> description of how cell is killed e.g. breaks open / makes holes, in cell surface membrane</p> <p><b>9</b> AVP ; e.g. proteins released from melanoma cell act as <u>non-self antigens</u></p> | <b>5</b> |